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## **Charged Lepton Mass Sum Rule from $U(3)$ -Family Higgs Potential Model**

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### **Abstract**

It is pointed out that if we assume  $U(3)$ -family nonet Higgs scalars which couple to charged leptons bilinearly, then a simple Higgs potential with a term which breaks the  $U(3)$ -family into  $SU(3)$ -family leads to a mass sum rule,  $m_e + m_\mu + m_\tau = (2/3)(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2$ . This sum rule is in excellent agreement with experiment. The sum rule is also in good agreement with experiment when applied to the masses of down-quarks,  $d$ ,  $s$  and  $b$ .

It is usually considered that the rapidly increasing mass pattern of the charged leptons ( $e, \mu, \tau$ ) [and also of up-quarks ( $u, c, t$ ) and down-quarks ( $d, s, b$ )] indicates that they do not belong to the same multiplet of particles, and cannot probably be treated by using the concept of “symmetry”. Therefore, this degree of freedom is usually called as “generations”. (However, there have also been attempts<sup>1</sup> which regard this degree of freedom as a symmetry, the so-called “horizontal symmetry”.)

Against this conventional picture, in this paper, it is pointed out that this mass pattern can be treated within the framework of symmetry, if we deal with the square-root masses rather than of the linear masses. Hereafter, we call “generations” as “families”, and pursue a symmetry approach.

We may think that if quarks and leptons are to be commonly governed by a fundamental law of physics, they will provide us some simple and pretty relations among their masses and mixings, and they may be derived from a relatively simple consideration.

In this paper, we show that a simple Higgs potential model can provide a well-satisfied charged-lepton mass sum rule within the framework of the  $SU(3)$ -family symmetry: We assume  $U(3)$ -family nonet Higgs scalars which couple to the charged leptons bilinearly<sup>2</sup>,

$$H_{lepton} = \frac{1}{F} \bar{e}^i \Phi_i^j \Phi_j^{\dagger k} e_k, \quad (1)$$

where  $i, j, k$  are family-indices and the coupling constant  $F$  has a dimension of mass. Then a very simple Higgs potential with a term which breaks the  $U(3)$  symmetry into an  $SU(3)$  symmetry leads to a mass sum rule<sup>3</sup>

$$m_e + m_\mu + m_\tau = \frac{2}{3} (\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2, \quad (2)$$

which is in excellent agreement with experiment. For the down-quarks, the corresponding sum rule to (2) is also in good agreement with experiment.

Our scenario which derives the sum rule (2) is very simple. We start from

the following Higgs scalar potential:

$$V = \frac{1}{2}\mu^2\text{Tr}[\Phi\Phi^\dagger] - \frac{1}{4}\lambda(\text{Tr}[\Phi\Phi^\dagger])^2 - \frac{1}{2}\lambda'\phi_0\phi_0^*\text{Tr}[\Phi_8\Phi_8^\dagger] , \quad (3)$$

where  $\Phi$ ,  $\Phi_8$  and  $\phi_0$  denote nonet scalars of  $U(3)$ -family, octet and singlet scalars of  $SU(3)$ -family, respectively:

$$\Phi = \Phi_8 + \frac{1}{\sqrt{3}}\phi_0\mathbf{1} , \quad (4)$$

$$\Phi_8 = \sum_{a=1}^8 \frac{1}{2}\lambda_a\phi_a , \quad (5)$$

$$\phi_0 = \frac{1}{\sqrt{3}}\text{Tr}[\Phi] . \quad (6)$$

The first two terms in (3) are the most popular expression of the  $U(3)$  invariant Higgs potential. The last term in (3) breaks the  $U(3)$ -family symmetry into  $SU(3)$ -family symmetry explicitly. Then the conditions  $\partial V/\partial\phi_0 = 0$  and  $\partial V/\partial\phi_a = 0$ , which are required for the minimum of the potential (3), gives

$$\mu^2 - \lambda\text{Tr}[\Phi\Phi^\dagger] - \lambda'\text{Tr}[\Phi_8\Phi_8^\dagger] = 0 , \quad (7)$$

and

$$\mu^2 - \lambda\text{Tr}[\Phi\Phi^\dagger] - \lambda'\phi_0\phi_0^* = 0 , \quad (8)$$

respectively, where we used the relation

$$\text{Tr}[\Phi\Phi^\dagger] = \text{Tr}[\Phi_8\Phi_8^\dagger] + \phi_0\phi_0^* . \quad (9)$$

Therefore the relations (7) and (8) leads to the relation

$$\text{Tr}[\Phi_8\Phi_8^\dagger] = \phi_0\phi_0^* = \frac{\mu^2}{2\lambda + \lambda'} , \quad (10)$$

i.e.,

$$\text{Tr}[\Phi\Phi^\dagger] = \frac{2}{3}\text{Tr}[\Phi]\text{Tr}[\Phi^\dagger] = \frac{2\mu^2}{2\lambda + \lambda'} , \quad (11)$$

because of  $\phi_0\phi_0^* = \text{Tr}[\Phi]\text{Tr}[\Phi^\dagger]/3$  and  $\text{Tr}[\Phi_8\Phi_8^\dagger] = \text{Tr}[\Phi\Phi^\dagger] - \text{Tr}[\Phi]\text{Tr}[\Phi^\dagger]/3$ . From the assumption (1), the charged lepton mass matrix  $M_{\text{down-lepton}}$  is given by the vacuum expectation values  $\langle \Phi \rangle$  of the nonet Higgs scalars  $\Phi$  as follows:

$$M_{\text{down-lepton}} \equiv \sqrt{M}\sqrt{M}^\dagger = \frac{1}{F} \langle \Phi \rangle \langle \Phi^\dagger \rangle . \quad (12)$$

Therefore, the relation (11) provides

$$\text{Tr}[\sqrt{M}\sqrt{M}^\dagger] = \frac{2}{3} \text{Tr}[\sqrt{M}]\text{Tr}[\sqrt{M}^\dagger] , \quad (13)$$

which gives the sum rule (2).

This sum rule is in excellent agreement with experiment<sup>4</sup>:

$$3\text{Tr}[M]/2(\text{Tr}[\sqrt{M}])^2 = 1.0006 \pm 0.0003 . \quad (14)$$

For down-quarks, the similar sum rule to (2) is also in good agreement with the observed down-quark mass values<sup>5</sup>  $m_d = 8.9 \pm 2.6 \text{ MeV}$ ,  $m_s = 175 \pm 55 \text{ MeV}$  and  $m_b = 5.3 \pm 0.1 \text{ GeV}$ :

$$3\text{Tr}[M]/2(\text{Tr}[\sqrt{M}])^2 = 1.038_{-0.039}^{+0.053} . \quad (15)$$

However, the similar sum rule for up-quark masses predicts a now too low top quark mass value,  $m_t \simeq 22 \text{ GeV}$ , from the observed values  $m_u \simeq 5.1 \text{ MeV}$  and  $m_c \simeq 1.35 \text{ GeV}$ . This suggests that the present scenario holds both for the down-leptons and down-quarks, but not for the up-quarks (also may be for up-leptons). For the time being, we consider our scenario only for the down-fermions.

We note that in the sum rule (2) the presence of the sizable value of the electron mass (also  $d$  quark mass in the down-quark mass sum rule) is essential for the excellent agreement with experiment. The sum rule (2) gives

$$\sqrt{\frac{m_\tau}{m_\mu}} = 2 + \sqrt{\frac{m_e}{m_\mu}} + \sqrt{3} \sqrt{1 + 4\sqrt{\frac{m_e}{m_\mu}} + \frac{m_e}{m_\mu}} , \quad (16)$$

so that if electron is exactly massless, then we predict somewhat smaller value of the ratio  $m_\tau/m_\mu$

$$\frac{m_\tau}{m_\mu} = \left( \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \right)^2 = (2 + \sqrt{3})^2 \approx 13.9 \quad (17)$$

compared with the experimental value  $m_\tau/m_\mu \approx 16.9$ .

There is an attractive proposal<sup>6</sup> which says that in the first step of the scenario electron should be exactly massless. We may regard that the predicted value (17) is in reasonable agreement with experiment and can be used as the first approximation in such scenario. However, as it stands, our above model does not possess any parameters which realize the massless electron.

If we want massless electron in our model, we must put some additional requirements. In our scenario, the values of  $\text{Tr}[\sqrt{M}]$  and  $\text{Tr}[(\sqrt{M})^2]$  are already fixed when the parameters  $\mu^2$ ,  $\lambda$  and  $\lambda'$  are once given, so that the next still-adjustable value is  $\text{Tr}[(\sqrt{M})^3]$ . Therefore, for example, let us add a requirement that the value  $\text{Tr}[(\sqrt{M})^3]$  should take as smaller value as possible. Then, using the general formula for any  $3 \times 3$  matrix  $A$

$$\det A = \frac{1}{3}\text{Tr}[A^3] - \frac{1}{2}\text{Tr}[A^2]\text{Tr}[A] + \frac{1}{6}(\text{Tr}[A])^3, \quad (18)$$

we obtain

$$\text{Tr}[(\sqrt{M})^3] = 3\det\sqrt{M} + \frac{1}{2}(\text{Tr}[\sqrt{M}])^3. \quad (19)$$

Since the second term in the right-hand-side of (19) is constant under the parameters  $\mu^2$ ,  $\lambda$  and  $\lambda'$  fixed, the requirement of the smallest  $\text{Tr}[(\sqrt{M})^3]$  leads to

$$\det\sqrt{M} = 0, \quad (20)$$

i.e.,  $m_e = 0$ . However, at present, the meaning of this additional requirement is unclear.

Although our sum rule (2) can be derived from the very simple potential model (3), there is a problem in the scenario, that is, in the assumption (1). We must explain why there are no linear-coupling terms  $\bar{e}\Phi e$  and  $\bar{e}\Phi^\dagger e$ . In order to forbid such linear-coupling terms, we must bring further additional symmetry or quantum number into the present scenario. For example, if we assume four nonet scalars,  $(2,1)+(1,2)$  of  $SU(2)_L \times SU(2)_R$ , we can make the bilinear-coupling term  $SU(2)_L \times SU(2)_R$  invariant and the linear-coupling terms forbidden.

Another idea which justifies the square-root mass formula is to assume heavy

leptons  $E$  with a common heavy mass value  $M$ :

$$H_{lepton} = f\bar{e}^i\Phi_i^j E_j + f^*\bar{E}^i\Phi_i^{\dagger j} e_j + M\bar{E}^i E_i . \quad (21)$$

Then, in a way similar to the so-called “seesaw” mechanism<sup>7</sup> in neutrino masses, the light lepton masses  $m(e_i)$  in the expression in which the vacuum expectation values  $\langle \Phi \rangle$  is diagonalized is given by

$$m(e_i) \simeq (fv_i)^2/M , \quad (22)$$

where  $v_i = \langle \Phi_i^i \rangle$ .

In any extension of the present scenario, the model will become somewhat more complicated. It is an open question how we can justify the bilinear-coupling term (1) still keeping the simplicity of the present scenario.

In conclusion, without discussing the details of the mass matrix form, we have obtained a down-fermion mass relation (2) which is in excellent agreement with experiment, from a simple Higgs potential of  $U(3)_{family}$  nonet scalars, where  $U(3)_{family}$  is explicitly broken into  $SU(3)_{family}$  and  $SU(3)_{family}$  is spontaneously broken. The present scenario may offer us a clue to the breaking pattern of the quark and lepton symmetry.

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